

EXACT NON-SCATTERING HOMOGENEOUS DIELECTRICS FROM SCHIFFER COUNTEREXAMPLES

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ABSTRACT. We prove the existence of bounded, simply connected, noncircular homogeneous isotropic dielectric inclusions that are exactly non-scattering for a nonzero physical incident field in the two-dimensional scalar Maxwell reduction. The incident field is the explicit normalized Herglotz wave

$$v(x) = \sqrt{2\pi} J_0(|x|), \quad h(d) = \frac{1}{\sqrt{2\pi}},$$

at exterior wavenumber one. The construction starts from the computer-assisted noncircular Schiffer counterexample of Colbrook and Stepaniants. We replace its constant boundary datum by a small-frequency Bessel wave, prove that the regular Schiffer zero persists, and certify the continuation uniformly for every $0 < \tau \leq 10^{-13}$. After dilation, the resulting domain and constant relative permittivity are

$$\Omega_\tau = \sqrt{\tau} \phi_{p_\tau}(\mathbb{D}), \quad q_\tau = \frac{p_{\tau,0}^2}{\tau}.$$

An interior Helmholtz field matches both Cauchy traces of v exactly, so the radiating scattered field and its far-field pattern vanish identically. As a consequence, every prescribed contrast $Q \geq 1.03 \times 10^{16}$ occurs for some noncircular member of the family.

1. INTRODUCTION

Let $D \subset \mathbb{R}^2$ be a bounded dielectric inclusion with constant relative permittivity $q > 0$ and constant magnetic permeability. In the out-of-plane-electric scalar reduction of time-harmonic Maxwell equations, an incident field v at exterior wavenumber $k > 0$ is physical when it is a nonzero entire solution of

$$(1) \quad (\Delta + k^2)v = 0 \quad \text{in } \mathbb{R}^2.$$

Exact non-scattering requires an interior field U satisfying

$$(2) \quad (\Delta + k^2q)U = 0 \quad \text{in } D$$

and the two trace identities

$$(3) \quad U = v, \quad \partial_\nu U = \partial_\nu v \quad \text{on } \partial D.$$

Then the exterior total field may equal v itself, so the scattered field is the zero radiating solution. This is stronger than the mere existence of a transmission eigenvalue: the incident component must extend to an entire physical Helmholtz solution, and in the present theorem it is an L^2 -Herglotz wave. The distinction is emphasized in [HV25, CV26].

Two lines of prior work frame this question. First, geometric obstructions: domains with corners scatter every incident wave [BPS14, PSV17, EH15], and non-scattering forces strong boundary regularity [SS21, CV23, CVX23], so any example must have smooth, indeed essentially

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real-analytic, boundary. Second, spectral necessity: non-scattering requires an interior transmission eigenpair [Kir86, CM88, CCH16], but a transmission eigenpair alone does not supply a physical incident field, and for broad families of perturbed discs the nonscattering wave numbers of physical incident waves are at most finite [VX21, HV25]. Radial homogeneous inclusions are the classical exception; whether any noncircular homogeneous inclusion admits an exactly non-scattering physical incident wave has remained open [CV26].

A neighboring overdetermined problem is Schiffer's conjecture, which is closely tied to the Pompeiu problem [Pom29, BST73, Wil76, Ber80]. It asks whether a bounded domain admitting a nonconstant solution of

$$(4) \quad (\Delta + \lambda)u = 0 \quad \text{in } D, \quad u = 1, \quad \partial_\nu u = 0 \quad \text{on } \partial D$$

must be a disc. Colbrook and Stepaniants recently constructed a computer-assisted noncircular counterexample with real-analytic boundary, and Cao-Labora and de Dios Pont independently produced infinitely many bifurcating counterexamples [CS26, CDP26]. These results do not immediately solve the scattering problem: at positive frequency the constant function 1 does not satisfy the exterior Helmholtz equation.

The central observation of this paper is that the Schiffer datum can be replaced by a genuine radial Helmholtz wave. More precisely, a regular Schiffer zero persists when the constant boundary datum is deformed to

$$J_0(\sqrt{\tau} |x|) = 1 - \frac{\tau}{4} |x|^2 + O(\tau^2).$$

The resulting transmission pair has exterior squared wavenumber τ and interior squared wavenumber close to the Schiffer eigenvalue. A final dilation fixes the exterior wavenumber at $k = 1$. This answers the constant-index, noncircular, physical-incident-field existence question affirmatively in the two-dimensional scalar model. Thus the construction is a subwavelength, high-contrast resonance: the physical diameter shrinks like $\sqrt{\tau}$, while the interior optical size remains finite.

1.1. Main result. Let

$$(5) \quad h(d) = \frac{1}{\sqrt{2\pi}}, \quad v(x) = \int_{\mathbb{S}^1} e^{ix \cdot d} h(d) \, ds(d) = \sqrt{2\pi} J_0(|x|).$$

Then $h \in L^2(\mathbb{S}^1)$, $\|h\|_{L^2} = 1$, and $(\Delta + 1)v = 0$ in $\mathbb{R}^2$.

Theorem 1.1 (Exact non-scattering family). *For every $0 < \tau \leq 10^{-13}$ there exist a bounded simply connected domain $\Omega_\tau \subset \mathbb{R}^2$, a constant $q_\tau > 0$, and an interior field U_τ such that:*

- (i) $\partial\Omega_\tau$ is a real-analytic Jordan curve and Ω_τ is not a disc;
- (ii) $q_\tau \neq 1$;
- (iii)

$$(\Delta + q_\tau)U_\tau = 0 \quad \text{in } \Omega_\tau;$$

- (iv) $U_\tau = v$ and $\partial_\nu U_\tau = \partial_\nu v$ on $\partial\Omega_\tau$.

Consequently the radiating scattered field and the far-field pattern vanish identically for the normalized Herglotz incident wave (5) at exterior wavenumber $k = 1$.

The construction is exact, not a finite multipole cancellation. For each τ , a contraction theorem selects a unique infinite coefficient pair (γ_τ, p_τ) in a fixed certified Banach ball. If

$$(6) \quad \phi_p(z) = z + \sum_{j \geq 1} \frac{p_j}{p_0(10j+1)} z^{10j+1},$$

then

$$(7) \quad \Omega_\tau = \sqrt{\tau} \phi_{p_\tau}(\mathbb{D}), \quad q_\tau = \frac{p_{\tau,0}^2}{\tau}.$$

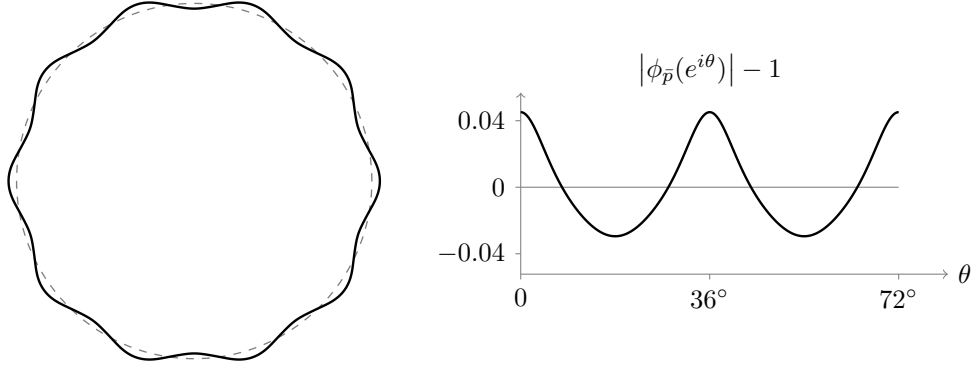


FIGURE 1. The certified Schiffer shape. Left: the image of the unit circle under $\phi_{\bar{p}}$ for the exact certificate center $\bar{p}$ of [CS26, CS26c] (solid), against the unit circle (dashed); the boundary radius varies between 0.9705 and 1.0452. Right: the radial profile $|\phi_{\bar{p}}(e^{i\theta})| - 1$ over two of the ten symmetry periods. Every domain $\Omega_\tau = \sqrt{\tau} \phi_{p_\tau}(\mathbb{D})$ of theorem 1.1 is a dilate of a small perturbation of this shape: the certified zero stays within 10^{-6} of $\bar{x}$ in the weighted norm.

Corollary 1.2 (Prescribed high contrast). *For every real $Q \geq 1.03 \times 10^{16}$ there exists a bounded simply connected non-disc with real-analytic Jordan boundary and constant relative permittivity exactly Q that is non-scattering for the same incident wave (5) at exterior wavenumber one.*

1.2. Proof mechanism. The argument has four steps.

- (1) The Colbrook–Stepaniants theorem supplies a regular noncircular Schiffer zero in a weighted coefficient algebra.
- (2) We introduce a Bessel-deformed fixed-disc equation $F_\tau = 0$ whose zero is exactly equivalent to matching the physical Bessel Cauchy data.
- (3) The implicit-function theorem gives a local branch abstractly, and explicit perturbation bounds show that the original contraction ball remains valid for all $0 \leq \tau \leq 10^{-13}$.
- (4) Conformal covariance and dilation produce the physical domain, constant contrast, and exact zero scattered field.

The full numerical provenance, the executable verifiers, and a Lean re-verification of the certificate arithmetic are separated into the accompanying verification supplement; the present article contains the mathematical construction and proof.

2. THE CERTIFIED SCHIFFER SEED

We use the fixed-disc formulation of [CS26]. Let X_ρ and Y_ρ be their weighted real coefficient spaces, with $\rho = 1.05$. A shape sequence has ten-fold symmetry,

$$(8) \quad p(z) = \sum_{j \geq 0} p_j z^{10j}, \quad p_0 > 0,$$

and is related to (6) by

$$(9) \quad p = p_0 \phi'_p.$$

The operator K is an explicit inverse of the Laplacian on the compatible coefficient range and satisfies

$$(10) \quad \Delta K\gamma = \gamma, \quad K\gamma = \partial_r K\gamma = 0 \quad \text{on } \partial\mathbb{D}.$$

The Schiffer equation is

$$(11) \quad F_0(\gamma, p) := \gamma + |p|^2(1 + K\gamma) = 0.$$

Indeed, if $W = K\gamma$ and $\tilde{u} = 1 + W$, then

$$\Delta \tilde{u} + |p|^2 \tilde{u} = 0 \quad \text{in } \mathbb{D}, \quad \tilde{u} = 1, \quad \partial_r \tilde{u} = 0 \quad \text{on } \partial \mathbb{D}.$$

Using (9), the push-forward to $\phi_p(\mathbb{D})$ has constant interior squared wavenumber p_0^2 .

The following summarizes precisely the consequences of the Colbrook–Stepanianis computer-assisted theorem used here.

Proposition 2.1 (Certified Schiffer seed). *There is a center $\bar{x} = (\bar{\gamma}, \bar{p}) \in X_\rho$, a radius $r = 10^{-6}$, an invertible bounded operator $\mathcal{A} : Y_\rho \rightarrow X_\rho$, and a unique exact zero $x_\star = (\gamma_\star, p_\star)$ of F_0 in the closed ball $B_r(\bar{x})$ with the following properties.*

- (a) *Let $T_0 = I - \mathcal{A}F_0$. The certificate provides exact cubic Taylor majorants $Y \leq 1.59 \times 10^{-10}$, $Z \leq 0.621$, $C_2 \leq 122$, $C_3 \leq 0.012$ such that, on $B_r(\bar{x})$,*

$$\sup_{B_r(\bar{x})} \|T_0(x) - \bar{x}\| \leq Y + Zr + C_2r^2 + C_3r^3, \quad \sup_{B_r(\bar{x})} \|DT_0\| \leq Z + 2C_2r + 3C_3r^2.$$

Writing $R_0 := Y + Zr + C_2r^2 + C_3r^3 - r$ for the invariant-ball defect and $L_0 := Z + 2C_2r + 3C_3r^2$ for the contraction constant, exact rational evaluation gives

$$(12) \quad R_0 \leq -0.000000378718999999988 < 0,$$

$$(13) \quad L_0 \leq 0.621244000000036 < 1,$$

so T_0 maps $B_r(\bar{x})$ into itself and is a strict contraction there.

- (b) *Every p in the ball satisfies*

$$(14) \quad \|p\|_\rho < 55, \quad 31 < p_0 < 32, \quad \sum_{j \geq 1} |p_j/p_0| < 0.65.$$

- (c) *ϕ_p is univalent on a neighborhood of $\overline{\mathbb{D}}$, and its first nonconstant shape coefficient does not vanish. Hence $\phi_p(\mathbb{D})$ is a bounded simply connected non-disc with real-analytic Jordan boundary.*

- (d) *The preconditioner satisfies*

$$(15) \quad \|\mathcal{A}\| < 1180.$$

Proposition 2.1 restates, in the notation of this paper, the certified outputs of [CS26, Eqs. (3)–(5) and (22), Lemma 2.4, Theorem 1.1] and of the released validation certificate [CS26c], whose pinned hashes and replay commands are recorded in the verification supplement. The bound (15) follows from the upstream identity $C_3 = \|\mathcal{A}\|/(96\bar{p}_0^2)$ together with $C_3 \leq 0.012$ and the exact center value $\bar{p}_0 < 32$:

$$\|\mathcal{A}\| = 96\bar{p}_0^2 C_3 < 96 \cdot 32^2 \cdot 0.012 = 1179.648 < 1180.$$

It therefore does not depend on reconstructing the norm of the upstream approximate inverse to many digits.

Remark 2.2 (Epistemic status). *Theorem 1.1 is conditional on the computer-assisted theorem of [CS26], which at the time of writing is a very recent preprint. Everything imported from it is confined to proposition 2.1; the accompanying release pins the upstream certificate by cryptographic hash, replays its authenticated checker, and machine-compares every constant in proposition 2.1 against the pinned upstream data.*

Lemma 2.3 (Regularity of the certified zero). *The derivative $DF_0(x_\star) : X_\rho \rightarrow Y_\rho$ is a bounded linear isomorphism.*

Proof. At the fixed point,

$$DT_0(x_\star) = I - \mathcal{A}DF_0(x_\star).$$

By (13), $\|DT_0(x_\star)\| < 1$. Hence $I - DT_0(x_\star) = \mathcal{A}DF_0(x_\star)$ is invertible by the Neumann-series lemma. Since $\mathcal{A}$ is invertible, so is $DF_0(x_\star)$. $\square$

3. THE BESSEL-DEFORMED TRANSMISSION EQUATION

For $\tau \geq 0$, define

$$(16) \quad H_{\tau,p}(z) := \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2} \left(\frac{\tau |\phi_p(z)|^2}{4} \right)^n.$$

For $\tau > 0$,

$$(17) \quad H_{\tau,p}(z) = J_0(\sqrt{\tau} |\phi_p(z)|),$$

and $H_{0,p} = 1$. The power-series definition makes the dependence on τ analytic at zero. Conformal covariance of the Laplacian and (9) give

$$(18) \quad \Delta H_{\tau,p} + \frac{\tau}{p_0^2} |p|^2 H_{\tau,p} = 0 \quad \text{in } \mathbb{D}.$$

We now define the deformed operator

$$(19) \quad F_\tau(\gamma, p) := \gamma + |p|^2 K\gamma + |p|^2 \left(1 - \frac{\tau}{p_0^2} \right) H_{\tau,p}.$$

At $\tau = 0$, this is exactly (11).

Lemma 3.1 (Fixed-disc transmission identity). *If $F_\tau(\gamma, p) = 0$ and*

$$(20) \quad \tilde{U}_{\tau,p} := H_{\tau,p} + K\gamma,$$

then

$$(21) \quad \Delta \tilde{U}_{\tau,p} + |p|^2 \tilde{U}_{\tau,p} = 0 \quad \text{in } \mathbb{D}.$$

Moreover, $\tilde{U}_{\tau,p}$ and $H_{\tau,p}$ have identical Dirichlet and radial Neumann traces on $\partial\mathbb{D}$.

Proof. Using (10) and (18), the left-hand side of (21) is exactly $F_\tau(\gamma, p)$. The trace statement follows from the two clamped trace identities in (10). $\square$

The preceding equation reveals the continuation mechanism independently of the particular numerical seed.

Theorem 3.2 (Regular Schiffer zeros generate physical non-scatterers). *Let $x_\star = (\gamma_\star, p_\star)$ be a zero of F_0 such that $DF_0(x_\star)$ is invertible, $p_{\star,0} > 0$, and $\phi_{p_\star}$ is univalent on a neighborhood of $\mathbb{D}$ and non-affine. Then there are $\varepsilon > 0$ and a real-analytic branch*

$$(-\varepsilon, \varepsilon) \ni \tau \mapsto x_\tau = (\gamma_\tau, p_\tau)$$

with $F_\tau(x_\tau) = 0$ and $x_\tau \rightarrow x_\star$ as $\tau \rightarrow 0$. For every sufficiently small $\tau > 0$, the scaled domain $\sqrt{\tau} \phi_{p_\tau}(\mathbb{D})$ is a noncircular homogeneous dielectric that is exactly non-scattering for the incident wave (5) at exterior wavenumber one.

Proof. The series (16) converges normally in a neighborhood of $(0, x_\star)$ in the weighted coefficient algebra. Hence $(\tau, x) \mapsto F_\tau(x)$ is real analytic there. The analytic implicit-function theorem and the invertibility of $DF_0(x_\star)$ produce the branch. Univalence on a complex collar and non-affineness are open in the weighted coefficient norm, so they persist for small τ . Non-affineness excludes a disc through the ten-fold symmetry of the ansatz (6): with $\omega = e^{2\pi i/10}$ one has $\phi_p(\omega z) = \omega \phi_p(z)$, so the image is invariant under rotation by ω ; a disc with this invariance is

centered at the origin, and the unique Riemann map of $\mathbb{D}$ onto such a disc with $\phi(0) = 0$ and $\phi'(0) > 0$ is linear. A non-affine ϕ_p therefore cannot map $\mathbb{D}$ onto a disc.

For $\tau > 0$, [lemma 3.1](#) supplies two fields with equal Cauchy data on the unit circle. Pushing them forward by ϕ_{p_τ} gives exterior squared wavenumber τ and interior squared wavenumber $p_{\tau,0}^2$. Dilation by $\sqrt{\tau}$ fixes the exterior wavenumber at one and changes the interior squared wavenumber to $p_{\tau,0}^2/\tau$. The complete physical verification is given in [section 5](#). $\square$

4. A UNIFORM CERTIFIED BRANCH

The abstract theorem gives a local branch but no explicit interval. We now show that the entire Schiffer contraction ball remains valid for $0 \leq \tau \leq 10^{-13}$. Put

$$\delta F_\tau := F_\tau - F_0.$$

All norms below are the weighted coefficient majorants of [\[CS26\]](#). One structural property of that algebra is used repeatedly and deserves to be isolated.

Lemma 4.1 (Radial multiplication is nonexpansive). *In the weighted coefficient algebra of [\[CS26, CS26c\]](#) the norm is a weighted ℓ^1 sum over a basis indexed by an angular block index ℓ and a radial index s , with weight $c_\ell \rho^\ell$ depending on ℓ only. Multiplication by the radial factor $|z|^2$ raises the radial index and leaves the angular index and the weight unchanged. Hence $\||z|^2 f\| \leq \|f\|$ for every f , and for any product decomposition $\phi_p(z) = z\psi_p(z)$,*

$$\||\phi_p|^2\| = \||z|^2 |\psi_p|^2\| \leq \|\psi_p\|^2.$$

Proof. The norm and the multiplication recurrences are specified in [\[CS26c, Validation Specification, §§2 and 4\]](#): the weights are $c_\ell \rho^\ell$ with $c_0 = 1$, $c_\ell = 2$, independent of the radial index, and multiplication by r^2 acts as the pure radial shift $(\ell, s) \mapsto (\ell, s+1)$. A shift with unchanged weights does not increase a weighted ℓ^1 norm, and the final inequality is submultiplicativity of the algebra norm. $\square$

Write $\phi_p(z) = z\psi_p(z)$. From [\(6\)](#) and [\(14\)](#),

$$(22) \quad \|\psi_p\| \leq 1 + \frac{1}{11p_0} \sum_{j \geq 1} |p_j| \rho^j \leq 1 + \frac{\|p\|_\rho - p_0}{11 \cdot 31} \leq 1 + \frac{55 - 31}{11 \cdot 31} < \frac{11}{10},$$

and therefore, by [lemma 4.1](#),

$$(23) \quad \||\phi_p|^2\| \leq \|\psi_p\|^2 < \frac{121}{100} < \frac{5}{4}.$$

Set $u = \tau \||\phi_p|^2\|/4$. For $0 \leq \tau \leq 10^{-13}$, $u < 1/2$. The full series [\(16\)](#), without truncation, therefore gives

$$(24) \quad \|H_{\tau,p}\| \leq 2, \quad \|H_{\tau,p} - 1\| \leq \frac{5}{8}\tau.$$

Since $p_0 > 31$,

$$(25) \quad \left\| \left(1 - \frac{\tau}{p_0^2}\right) H_{\tau,p} - 1 \right\| \leq \|H_{\tau,p} - 1\| + \frac{\tau}{p_0^2} \|H_{\tau,p}\| < \frac{63}{100}\tau.$$

Multiplication by $\|p\|^2 < 55^2$ yields

$$(26) \quad \boxed{\|\delta F_\tau\| < 1906\tau.}$$

For the derivative estimate, let ξ be a shape variation of full X_ρ norm at most one. The upstream product norm weights the shape block by $2\bar{p}_0\rho^j$ [CS26c, Validation Specification, Eq. (13)], so

$$(27) \quad \|\xi\| \leq \frac{1}{2\bar{p}_0} < \frac{1}{62}, \quad |\xi_0| < \frac{1}{62}.$$

Differentiating (6) and using (14) gives

$$(28) \quad \|D\psi_p[\xi]\| \leq \frac{1}{11} \left(\frac{\|\xi\|}{31} + \frac{55|\xi_0|}{31^2} \right) < \frac{1}{7000}.$$

Consequently

$$(29) \quad \|D|\phi_p|^2[\xi]\| \leq 2\|\psi_p\|\|D\psi_p[\xi]\| < \frac{1}{3000}, \quad \|DH_{\tau,p}[\xi]\| \leq \frac{\tau}{3000},$$

where the first estimate again uses lemma 4.1. Differentiating (19), and combining (25), (24), (27), and (29), gives

$$(30) \quad \boxed{\|D\delta F_\tau\| < 3\tau.}$$

The elementary constant arithmetic behind (26) and (30) is recorded in the verification supplement.

Define

$$T_\tau(x) = x - \mathcal{A}F_\tau(x).$$

On the original radius- r ball, (15), (26), and (30) give

$$(31) \quad R_\tau \leq R_0 + 1180 \cdot 1906 \tau,$$

$$(32) \quad L_\tau \leq L_0 + 1180 \cdot 3 \tau.$$

At $\tau_* = 10^{-13}$,

$$(33) \quad R_{\tau_*} \leq -0.00000015381099999998 < 0,$$

$$(34) \quad L_{\tau_*} \leq 0.621244000354036 < 1.$$

The right-hand sides of (31)–(32) increase with τ . Hence T_τ maps the same closed ball into itself and is a strict contraction for every $0 \leq \tau \leq \tau_*$. Banach's theorem gives a unique exact zero

$$(35) \quad x_\tau = (\gamma_\tau, p_\tau) \in B_r(\bar{x}) \quad (0 \leq \tau \leq 10^{-13}).$$

Because the whole branch stays in the certified ball, all univalence, analytic-boundary, and noncircularity conclusions of proposition 2.1 hold uniformly.

5. THE PHYSICAL INCLUSION AND EXACT ZERO SCATTERING

Fix $0 < \tau \leq 10^{-13}$ and the exact zero $x_\tau = (\gamma_\tau, p_\tau)$ from (35). Define the domain and contrast by (7). For $y \in \Omega_\tau$, put

$$z = \phi_{p_\tau}^{-1} \left(\frac{y}{\sqrt{\tau}} \right)$$

and define

$$(36) \quad U_\tau(y) = \sqrt{2\pi} [H_{\tau,p_\tau}(z) + K\gamma_\tau(z)].$$

Since $y = \sqrt{\tau} \phi_{p_\tau}(z)$,

$$(37) \quad H_{\tau,p_\tau}(z) = J_0(|y|).$$

Conformal covariance transforms (21) into an interior Helmholtz equation with squared wavenumber $p_{\tau,0}^2$ on $\phi_{p_\tau}(\mathbb{D})$. The dilation $y = \sqrt{\tau}x$ therefore gives

$$(38) \quad (\Delta_y + q_\tau)U_\tau = 0 \quad \text{in } \Omega_\tau, \quad q_\tau = \frac{p_{\tau,0}^2}{\tau}.$$

By (14),

$$(39) \quad q_\tau > \frac{31^2}{10^{-13}} > 1,$$

so q_τ is positive and nonunit.

It remains to check the traces. The difference between the pulled-back interior field and the pulled-back incident field is $K\gamma_\tau$, whose Dirichlet and radial Neumann traces vanish by (10). Under a conformal map the normal derivative is multiplied by a nonzero boundary scale factor, and under dilation it is multiplied by another nonzero constant. Thus zero remains zero, and (37) gives

$$(40) \quad U_\tau = v, \quad \partial_\nu U_\tau = \partial_\nu v \quad \text{on } \partial\Omega_\tau.$$

For completeness, writing $d = (\cos \theta, \sin \theta)$,

$$\int_0^{2\pi} e^{ix \cdot d} d\theta = 2\pi J_0(|x|),$$

so (5) follows, and

$$\|h\|_{L^2(\mathbb{S}^1)}^2 = \int_0^{2\pi} \frac{1}{2\pi} d\theta = 1.$$

The incident field is therefore a nonzero normalized L^2 -Herglotz wave.

Define the total field piecewise by U_τ in Ω_τ and by v in the exterior. Equation (40) makes it a transmission solution. Its exterior scattered component is identically zero, which satisfies the Sommerfeld radiation condition and has zero far-field pattern. Because q_τ is a bounded positive constant, the direct transmission problem for the incident field v has exactly one solution [CK19, Chapter 8]; the solution just exhibited is therefore the physical total field, and the scattered field of Ω_τ for v vanishes identically. This proves theorem 1.1.

6. CONSEQUENCES AND INTERPRETATION

6.1. Prescribing the contrast. The contraction maps T_τ depend continuously on τ and have a uniform Lipschitz constant strictly below one. Hence their unique fixed points depend continuously on τ . One may see this directly from

$$(41) \quad \|x_\tau - x_\sigma\| \leq \frac{\|T_\tau(x_\sigma) - T_\sigma(x_\sigma)\|}{1 - L},$$

where $L < 1$ is any common contraction constant. Thus $\tau \mapsto p_{\tau,0}$ and

$$q_\tau = \frac{p_{\tau,0}^2}{\tau}$$

are continuous on $(0, 10^{-13}]$. Since $p_{\tau,0} \rightarrow p_{*,0} > 31$ as $\tau \downarrow 0$,

$$(42) \quad q_\tau \longrightarrow +\infty.$$

At $\tau_* = 10^{-13}$, the upper bound $p_{\tau_*,0} < 32$ gives

$$(43) \quad q_{\tau_*} < \frac{32^2}{10^{-13}} = 1.024 \times 10^{16}.$$

For any $Q \geq 1.03 \times 10^{16}$, choose $\delta > 0$ small enough that $q_\delta > Q$. The intermediate-value theorem on $[\delta, \tau_*]$ and (43) produce some τ with $q_\tau = Q$. This proves [corollary 1.2](#).

6.2. The subwavelength/high-index regime. At fixed exterior wavenumber one, the diameter of Ω_τ is $O(\sqrt{\tau})$, whereas

$$(44) \quad \sqrt{q_\tau} \sqrt{\tau} = p_{\tau,0} \longrightarrow p_{*,0} \approx 31.967.$$

Thus the family maintains a finite interior optical size by increasing the positive refractive index as the physical inclusion shrinks. The result is an existence theorem for homogeneous, passive, positive-index material, but it does not yet reach moderate contrast. Natural next questions are whether other Schiffer branches, nonradial Herglotz controls, or continuation farther from $\tau = 0$ can reduce the required contrast, and whether an analogous mechanism exists for the full three-dimensional vector Maxwell system.

REPRODUCIBILITY AND FORMAL VERIFICATION

The accompanying release contains the exact certificate data, independent exact-arithmetic verifiers that rederive the perturbation constants of [section 4](#) from the certified ball bounds, the pinned and replayed upstream dependency, and a Lean 4 re-verification of the exact rational certificate arithmetic together with generic fixed-point lemmas in the form used here. The Lean development does not formalize the operator estimates, the upstream certificate, or the scattering analysis; its exact boundary, the numerical provenance, and the replay commands are recorded in a separate verification supplement so that they do not obscure the mathematical argument above. The complete release is archived at Zenodo, [doi:10.5281/zenodo.22016374](https://doi.org/10.5281/zenodo.22016374), and developed at <https://github.com/Mansehej/non-scattering-schiffer>.

DECLARATION OF COMPUTATIONAL ASSISTANCE

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